

Omnivortex V1 — Phase 1 Measurement Protocol

R=89mm r=55mm · 123 turns · 22 AWG · 3.0Ω DC | github.com/Te55er4ct/phi-harmonic-toroidal-field

Pre-measurement environment check

Laptop on battery · WiFi off · Wooden sawhorse bench · Confirm DC resistance = 3.0Ω before any RF measurement

Step 1 LCR meter: Inductance and Q factor

1a Frequency set

Measure at 1 kHz
then 10 kHz
then 100 kHz
Record each reading

1b Record L and Q

Log inductance (μH or mH)
Log Q factor
Log ESR (Ω) if displayed
Note drift over 30 s

1c Expected range

L ~ 0.8–3 mH typical
Q may drop above 10 kHz
Flag if Q > 100 or < 5
Verify R_DC matches 3.0 Ohm

Step 2 A+H NanoVNA: self-resonant frequency (SRF)

2a Calibrate NanoVNA

Perform OSL (Open / Short / Load)
Use same SMA pigtails as measurement
Span: 100 kHz → 30 MHz
Port 1 (CH0) only for SRF scan

2b Locate SRF peak

Plot |Z| or S11 on screen
SRF = frequency at |Z| maximum
Coil acts capacitive above SRF
Log exact frequency in MHz

SRF cross-check formula

$SRF = 1 / (2 \times \pi \times \sqrt{L \times C_{self}})$ — use measured L to back-calculate C_self after SRF is known

Step 3 A+H NanoVNA: impedance sweep 100 kHz → 30 MHz

3a Setup sweep

Start: 100 kHz / Stop: 30 MHz
Points: 101 minimum
Display: |Z| and Phase
Also capture Smith chart view

3b Key features

Inductive rise at low frequencies
SRF peak = |Z| maximum
Note each anti-resonance dip
Phase zero-crossings: log freq

3c Data export

Save .s1p Touchstone file
Screenshot full sweep
Repeat twice, compare traces
Log ambient temp and time

Post-measurement calculations

Wheeler toroid: $L = \mu_0 \times N^2 \times A / (2 \times \pi \times R)$ where $A = \pi \times r^2$
 $A = \pi \times (0.055)^2 = 9.503e-3 \text{ m}^2$ $R = 0.089 \text{ m}$ $N = 123$
 $L_{\text{predicted}} \sim \mu_0 \times 15129 \times 9.503e-3 / (2 \times \pi \times 0.089) \sim 3.23 \text{ mH}$ (expect 5-15% lower measured)
C_self (from SRF): $C = 1 / (4 \times \pi^2 \times f_{\text{SRF}}^2 \times L)$ $Q = \omega \times L / R_{\text{ESR}}$

Omnivortex V1 — Measurement Log Sheet

One sheet per measurement session · R=89mm r=55mm · 123T · 22 AWG

Date: _____ Time: _____ Temp (°C/F): _____ Humidity %: _____ Location: _____

Section 1 — LCR Meter (Proster)

Parameter	1 kHz	10 kHz	100 kHz	Units	Notes
Inductance L				mH	
Q factor				—	
ESR (R _s)				Ω	
DC resistance	—	—	—	Ω	Multimeter

Section 2 — NanoVNA Self-Resonant Frequency

Measurement	Value	Units	Notes
SRF (primary peak)		MHz	
SRF (harmonic 2)		MHz	
SRF (harmonic 3)		MHz	
C _{self} (calculated)		pF	$C = 1/(4\pi^2 f^2 L)$

Section 3 — Impedance Sweep Key Points (100 kHz → 30 MHz)

Feature	Freq (MHz)	Z (Ω)	Phase (°)	Notes
Inductive region start				
SRF (primary)				
Anti-resonance 1				
Anti-resonance 2				
Anti-resonance 3				
Phase zero-crossing 1				
Phase zero-crossing 2				
Capacitive cutoff				

Section 4 — Instrument & File Log

NanoVNA calibration file: _____

Touchstone .s1p filename: _____

Screenshot filename: _____

NanoVNA firmware version: _____

LCR meter model / serial: _____

Observations / anomalies / environmental notes

Operator: _____ Session #: _____ Git commit hash: _____